sEMG-Based Gesture Recognition Using Temporal History

Chaerin Hong, Seongsik Park, *Member, IEEE*, and Keehoon Kim, *Member, IEEE*

***Abstract*—Surface electromyography (sEMG) patterns have been decoded using learning-based methods that determine complicated nonlinear decision boundaries. However, overlapping classes in sEMG pattern recognition still degrade the classification accuracy because they cannot be separated by the decision boundaries. We hypothesized that certain overlapping classes can be separated while tracing the temporal history of sEMG patterns. Therefore, a novel post-processing method is proposed to adjust classification errors using the separated patterns from the temporal history of overlapping classes. The proposed method confirms the confidence of the prediction result by calculating the instantaneous pattern separability for the sequential sEMG input. The prediction result with high separability pattern is considered to have a high confidence of being correct (reliable). This result is stored for adjusting the next sEMG input. When the subsequent prediction is identified as having low confidence (unreliable), the predicted result is adjusted using the stored reliable predicted results. The proposed method adds an adjustment step to an existing classifier (maximum likelihood classifier (MLC), k-nearest neighbor (KNN), and support vector machine (SVM)), such that it can be attached to the back-end regardless of the type of classifier. Ten subjects performed 13 types of hand gestures, including overlapping patterns. The overall classification accuracy was enhanced to 88.93%(+8.12%p, MLC), 91.31%(+7.68%p, KNN), and 99.65%(+11.63%p, SVM) after the implementation of the proposed post-processing. Additionally, a faster and more accurate gesture classification was achieved with accuracy enhancement before gesture completion as 85.62%(+4.23%p, MLC), 89.77%(+4.23%p, KNN), and 97.62%(+11.12%p, SVM).**

***Index Terms*—Surface electromyography (sEMG), pattern recognition, transient analysis, post-processing, sequence, gesture recognition, temporal history.**

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Chaerin Hong is with the Department of Mechanical Engineering, Pohang University of Science and Technology (POSTECH), South Korea.

Seongsik Park is with the School of AI Convergence, Dongguk University, South Korea.

Keehoon Kim is with the Department of Mechanical Engineering, Pohang University of Science and Technology (POSTECH) Gyeongbuk 37673, South Korea, and also with the Institute for Convergence Research and Education in Advanced Technology, Yonsei University, 50 Yonsei-ro Seoul 03722, Republic of Korea (e-mail: khk@postech.ac.kr).

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I. Introduction

SURFACE electromyography (sEMG) pattern recognition has been proposed for hand-gesture recognition to decode user intentions in prosthesis control or human-robot interaction because the sEMG pattern contains muscle activation of a hand gesture [1], [2], [3], [4], [5], [6], [7], [8]. Advanced learning-based pattern recognition approaches have enhanced classification accuracy by finding complicated nonlinear decision boundaries among various sEMG patterns. However, certain sEMG patterns overlap each other even when they are generated from different gestures (overlapping class) because of the complex characteristics of sEMG signals (e.g., musculoskeletal system, crosstalk) [9]. These overlapping sEMG patterns degrade the classification accuracy because nonlinear decision boundaries cannot distinguish them. Therefore, overlapping sEMG patterns remain a fundamental challenge in sEMG-based gesture recognition.

Overlapping patterns have been considered in previous studies to achieve better performance. Some notable approaches include feature engineering and post-processing. Feature engineering finds sEMG features to improve the separability of sEMG patterns, which determine the decision boundaries between overlapping patterns [6], [10], [11]. Post-processing rejects or adjusts classification results (errors) in overlapping patterns by evaluating the confidence [12], [13], [14] or voting for the majority [1], [15]. However, these studies handled overlapping patterns with limited information because only sEMG signals in the *steady state* were used. The steady state represents partial information in the sEMG sequence of one gesture (Fig. 1(a)) and contains fewer temporal characteristics even though the sEMG signals are sequential data (time series) [16], [17]. Thus, we hypothesized that the overlapping patterns could be decoded better if other properties of the sequential sEMG data were added.

We used the sequential (temporal) characteristic of sEMG signals, including the *transient state* information, for the following reasons: First, sequential models can observe the temporal characteristics of sequential sEMG data with high accuracy [18], [19]. Second, transient state signals can enhance the accuracy because the transient state contains more

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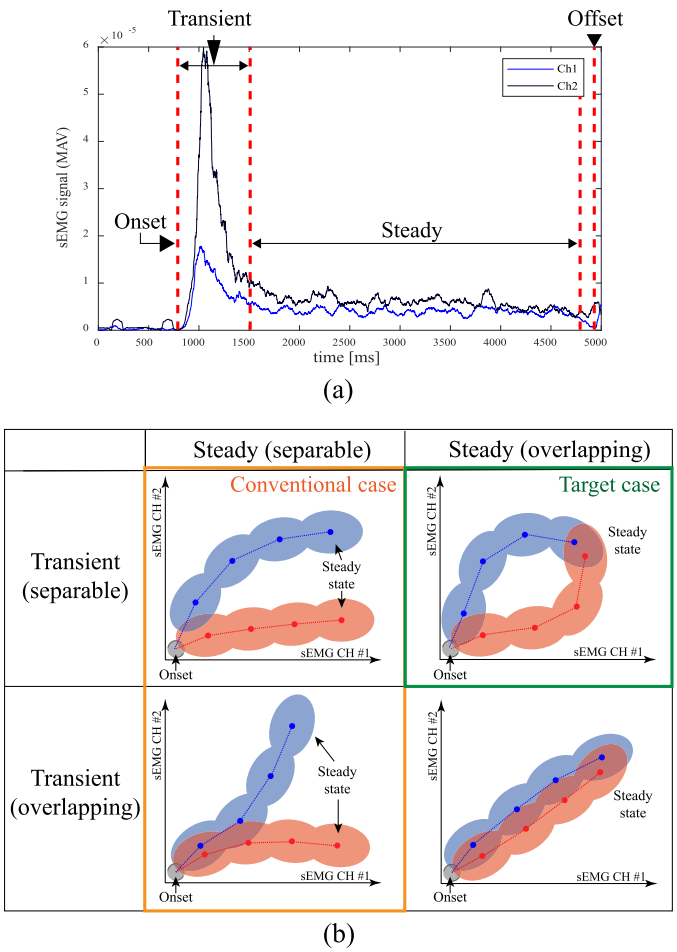

Fig. 1. (a) sEMG sequence (one gesture repetition) in which a mean absolute value (MAV) was applied. The sequence starts from the onset point, fluctuates in the transient state, and stabilizes in the steady state. (b) Representative types of two sEMG sequence pairs appear when observing the sEMG sequence in the sEMG signal channel space. Conventional studies have only focused on the separable steady state signals (upper and lower-left case). Our target problem is overlapping classes in the steady state along with separated patterns in the transient state, given in the upper right case.

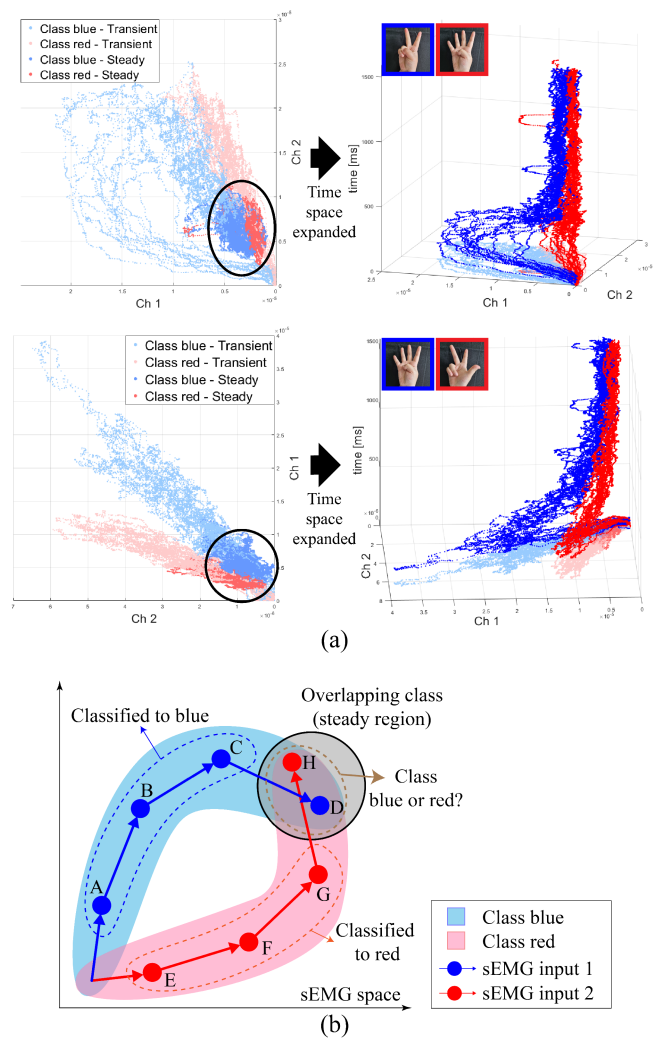

Fig. 2. (a) Examples of sEMG sequence pairs with separable patterns in a transient state and overlapped patterns in a steady state (upper right in Fig. 1(b)). (b) The sEMG inputs (D and H) near the overlapping sEMG patterns have a high probability of being erroneous. However, the classification result of those inputs can be adjusted using their temporal history (A-B-C for D, E-F-G for H) by referring to the classified results of the past sequential inputs.

information about temporal characteristics than the steady state [20], [21]. Additionally, separable patterns appeared in the transient state when observing the temporal history of overlapping patterns in the steady state, as shown in the upper right of Fig. 1(b) and Fig. 2(a) [22]. Although quantifying and analyzing transient state signals is challenging because of their low repeatability, researchers have demonstrated the usability of transient state signals, especially for dexterous control [23], [24]. In the same context, we exploit the sEMG sequence, including the transient state, to solve the overlapping sEMG pattern problem owing to (i) the possibility of being separated even overlapping in the steady state signals and (ii) faster classification than the case that only uses steady state signals.

In this paper, we present a novel post-processing method that adjusts the classification results (errors), particularly for overlapping patterns (Fig. 2(b)). The main idea of the proposed method is to identify and store the classification results from the separated patterns of the sEMG sequence for every sEMG input, from its onset (*reliability test*). The stored results are then used to adjust the classified results of the subsequent sEMG input if the input is classified into overlapping patterns (*self-adjustment*). Thus, there are two main advantages of the proposed method.

1) It is a novel post-processing framework that can be implemented for various conventional classifiers to enhance classification accuracy.
2) It makes a faster decision before the gesture motion is completed with enhanced accuracy.

The effect of the proposed algorithm is verified with experiments using 13 types of gestures, including gestures that exhibit overlapping sEMG patterns.

The paper is organized as follows. Section II illustrates the proposed post-process, defines some metrics for analyzing results, and Section III describes the experimental setup. Section IV presents the results with a discussion. The first experimental

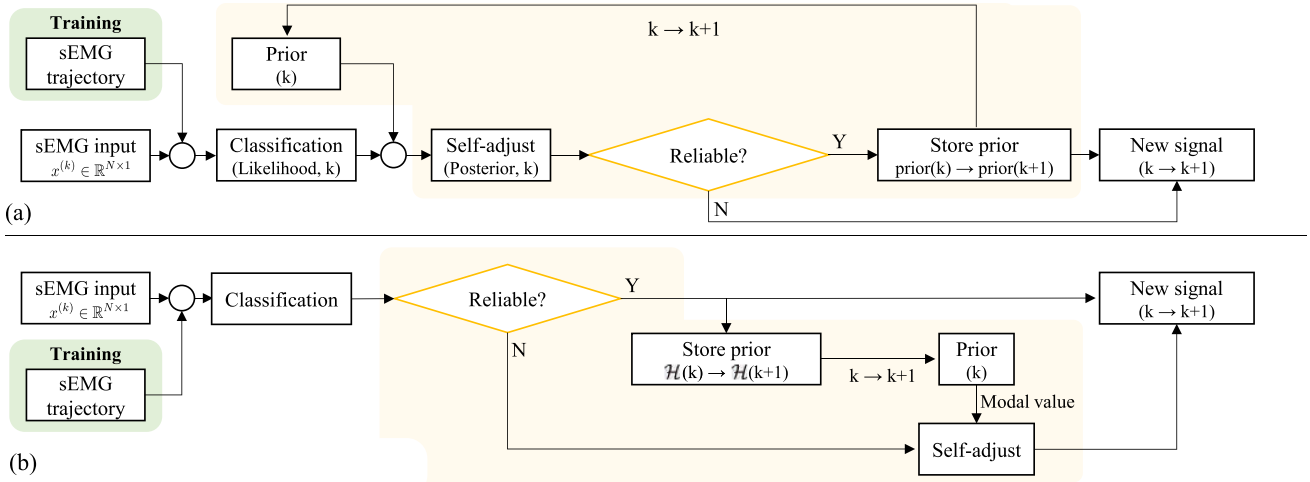

Fig. 3. Overall classification process with the suggested post-processing (box highlighted in yellow). The classification of the sequential sEMG input starts from the onset of an sEMG signal with a trained sEMG trajectory. Subsequently, the post-processing adjusts unreliable classification results with reliable classification results from the previous sequential inputs. In this paper, the post-processing implementation is illustrated with three classifiers: (a) probability-based classifier (MLC) and (b) distance-based classifiers (KNN and SVM).

result shows the existence of overlapping classes in our experiment and gives the correlation between overlapping classes and classification accuracy. Then, the second result compares the classification accuracy after implementing the proposed post-process in both steady and transient states. Section V provides concluding remarks for the entire study.

II. Methods

This section introduces the proposed post-processing method for adjusting classification errors in overlapping patterns. Section II-A defines the *sEMG trajectory* that provides a time-scale expansion to conventional classifiers for observing the sequential (temporal) property of sEMG signals. Then, a detailed explanation of the proposed method follows i) identify separable patterns for obtaining reliable results (*reliability test*), and ii) adjust errors in the overlapping classes (*self-adjustment*). Notably, this paper proposes a post-processing algorithm that can be coupled with conventional classifiers as Fig. 3. Thus, the overall steps and implementation of the proposed post-processing are illustrated in Section II-B to II-D with examples of three conventional classifiers (maximum likelihood classifier (MLC), k-nearest neighbor (KNN), and support vector machine (SVM)).

The following sections propose metrics for analyzing overlapping patterns and classification accuracy. The degree of overlap is defined to quantify how much two patterns are actually overlapping. Then, the calculation methods for classification accuracy are described.

A. Time-Scale Expansion: SEMG Trajectory

The sEMG sequence $\mathbf{x}_{\mathrm{c,i}}$ which mean absolute value (MAV) was applied is denoted as

$$\mathbf{x}_{\mathrm{c,i}} = \left[x_{\mathrm{c,i}}^{(1)}, x_{\mathrm{c,i}}^{(2)}, \ldots, x_{\mathrm{c,i}}^{(\mathrm{k})}, \ldots, x_{\mathrm{c,i}}^{(\mathrm{K})}\right] \in \mathbb{R}^{\mathrm{N}\times\mathrm{K}}, \quad (1)$$

where subscript $(\cdot)_{\mathrm{c}}$ indicates the labeled class and subscript $(\cdot)_{\mathrm{i}}$ indicates the gesture repetition index. The superscript $(\cdot)^{(k)}$ denotes the kth time step in the sequential sEMG data. The onset point of the sequence is denoted as $k = 1$, whereas $k = \mathrm{K}$ denotes the endpoint of the sequential sEMG data.

The proposed method exploits the classification results in the temporarily separable patterns of the transient state to adjust errors in the overlapping classes. However, some classifiers' classification rules are unsuitable for capturing the separable patterns because the classifiers only treat the spatial information of the sEMG signal. For example, MLC assigns one distribution for each class as shown in the left-hand illustration of Fig. 4(a) even if the transient state patterns are separable, whereas the steady state patterns are overlapped. Thus, we introduced the concept of *sEMG trajectory* to track the time-varying transient state sensitively, and ii) to find temporarily separable patterns in the sEMG sequence.

The sEMG trajectory is a gathering of parameters that reflect the status of the sEMG sequence at a certain time step (instantaneous pattern) as shown in the right-hand illustration of Fig. 4(b). In other words, the sEMG trajectory treats the sEMG signals at the sequential data level by storing the status of the signals for each time step. The parameters of the sEMG trajectory differ from the type of classifier (e.g., MLC, KNN). If the conventional classifier can capture the separability of a certain time step (temporal property), then the sEMG trajectory need not be assigned. The details for assigning the sEMG trajectory are described in subsequent sections.

B. Case 1: Maximum Likelihood Classifier (MLC)

1) sEMG Trajectory: MLC can be extended to the sEMG trajectory by assigning multivariate Gaussian distributions to each time step. In other words, the sEMG trajectory of MLC is similar to a regression line of the sequence data in the training set,

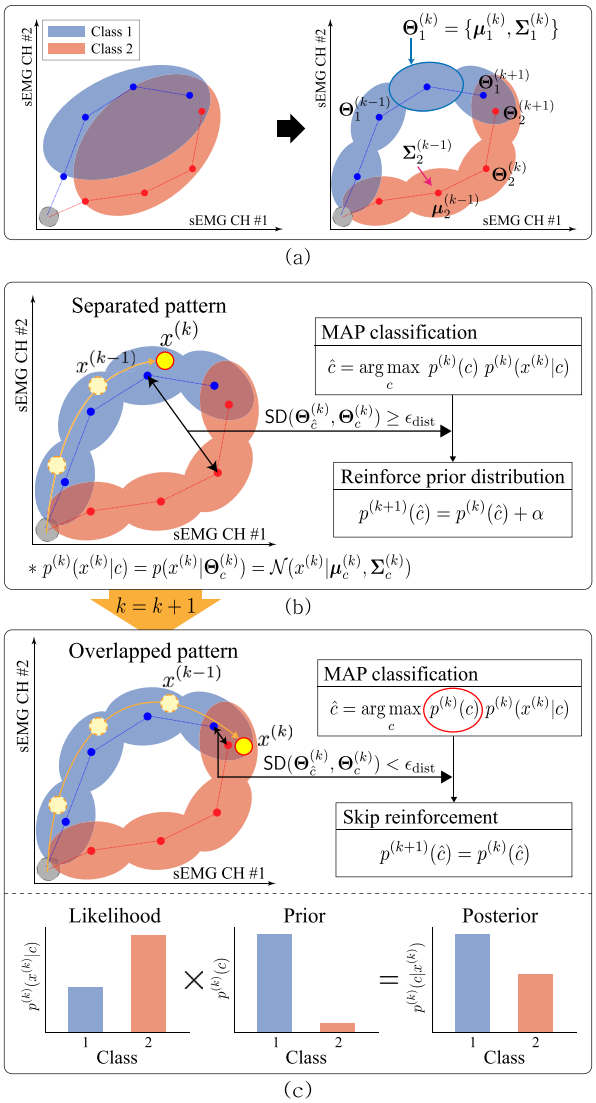

Fig. 4. (a) Time-scale expansion for MLC using the concept of sEMG trajectory. (b) Classification in the separable patterns. The current sEMG input $x^{(k)}$, which has the maximum likelihood and posterior probability value, is classified as Class 1. The statistical distance (SD) between the likelihood probability distribution indicates that they are separable in the kth time step; therefore, the classification results of $x^{(k)}$ are reliable (predicted in the separated patterns). Thus, the prior probability of Class 1, which is used for the self-adjustment step, is reinforced. (c) Classification and post-processing in the subsequent time step. The current input gives unreliable classification results (predicted in the overlapping patterns) because it is inside the overlapped probability distribution. However, a prior distribution that reflects the reliable results in the past sequential input adjusts the classification results by providing bias to reliable classification results (class 1).

which is represented as a trail of Gaussian distributions, as shown in Fig. 4(a). Thus, the parameters of the sEMG trajectory are the mean and covariance of the multivariate Gaussian distribution.

- sEMG trajectory $\boldsymbol{\Theta}_c$: The sEMG trajectory of the kth time step for MLC is a tuple of two parameters as

$$\boldsymbol{\Theta}_c^{(k)} = (\boldsymbol{\mu}_c^{(k)}, \boldsymbol{\Sigma}_c^{(k)}). \tag{2}$$

The mean ($\boldsymbol{\mu}_c^{(k)} \in \mathbb{R}^{N\times 1}$) is the average of the sEMG signals near k^{th} time step as

$$\boldsymbol{\mu}_c^{(k)} = \frac{\sum_{i\in I_{\text{TRN}}} \sum_{t=k_s}^{k_e} x_{c,i}^{(t)}}{(k_e - k_s)\cdot |I_{\text{TRN}}|}, \tag{3}$$

with time interval $w \in \mathbb{N}$, where $k_s = k - (w/2)$ and $k_e = k + (w/2)$. I_{TRN} denotes the repetition indices for registering (training) the sEMG trajectory.

The covariance ($\boldsymbol{\Sigma}_c^{(k)} \in \mathbb{R}^{N\times N}$) reflects the direction of the sEMG trajectory progression and variability of sEMG signals in k^{th} time step. The predetermined diagonal matrix $\boldsymbol{\Sigma}_{\text{pre}}$ was aligned with the direction of trajectory progression with linear transformation matrix $\mathbf{U}$ after finding the $\mathbf{U}$ by conducting singular value decomposition (SVD) on the sample covariance of the sEMG signals near k^{th} time step as

$$\begin{aligned} \mathbf{S}_c^{(k)} &= \sum_{i\in I_{\text{TRN}}} \sum_{t=k_s}^{k_e} \frac{(x_{c,i}^{(t)} - \boldsymbol{\mu}_c^{(k)})(x_{c,i}^{(t)} - \boldsymbol{\mu}_c^{(k)})^\top}{(k_e - k_s)\cdot |I_{\text{TRN}}| - 1}, \\ \mathbf{U}\boldsymbol{\Sigma}_{\text{SVD}}\mathbf{V} &= \mathbf{S}_c^{(k)}, \\ \boldsymbol{\Sigma}_c^{(k)} &= \mathbf{U}\boldsymbol{\Sigma}_{\text{pre}}\mathbf{U}^\top. \end{aligned} \tag{4}$$

As alluded to above, the status of the sEMG sequence in the kth time step is represented by a multivariate Gaussian distribution with mean $\boldsymbol{\mu}_c^{(k)}$ and covariance $\boldsymbol{\Sigma}_c^{(k)}$; therefore, the overall representation of the sEMG trajectory $\boldsymbol{\Theta}_c$ is as follows:

$$\boldsymbol{\Theta}_c = (\boldsymbol{\Theta}_c^{(1)}, \boldsymbol{\Theta}_c^{(2)}, \ldots, \boldsymbol{\Theta}_c^{(k)}, \ldots, \boldsymbol{\Theta}_c^{(K)}). \tag{5}$$

- Classification using the sEMG trajectory: The classification using the sEMG trajectory is similar to the MLC, which determines the class that gives the maximum likelihood value. The only difference is that the sEMG trajectory uses multivariate Gaussian distributions for a certain time step when classifying the new k^{th} sEMG input as

$$p^{(k)}(x^{(k)}|c) = p(x^{(k)}|\boldsymbol{\Theta}_c^{(k)}) = \mathcal{N}(x^{(k)}|\boldsymbol{\mu}_c^{(k)}, \boldsymbol{\Sigma}_c^{(k)}). \tag{6}$$

and finds class c to have the maximum likelihood value; therefore, the classification result $\hat{c}$ in the kth time step is shown as

$$\hat{c} \leftarrow \arg\ \max_c p^{(k)}(x^{(k)}|c). \tag{7}$$

Classification using sEMG trajectory can capture the patterns of both transient and steady states even if their patterns differ because the classification is conducted using the likelihood probability distribution in a certain time step. However, incorrect classification still occurs if the pattern in a certain time step overlaps with another pattern.

2) Post-Processing: The proposed post-processing is composed of two parts: *reliability test* (Fig. 4(b)) and *self-adjustment* (Fig. 4(c)) as alluded to above.

- *Reliability test:* We defined a *reliable* result as a classified result of sEMG input $x^{(k)}$ that was classified in the

Algorithm 1: Post-Processing Implementation in MLC.

Initialize:
  $p^{(1)}(c) \leftarrow \mathsf{unif}\{1, C\}$
for $1 \leq k \leq K$ **do**
  $x^{(k)} \leftarrow$ collectSample();
  for $1 \leq c \leq \mathrm{C}$ **do**
    $p^{(k)}(c|x^{(k)}) \leftarrow p^{(k)}(c)p^{(k)}(x^{(k)}|c)$
  end for
  $\hat{c} \leftarrow \arg\ \max_c p^{(k)}(c|x^{(k)})$
  if $\min_c(\mathsf{SD}(\boldsymbol{\Theta}_{\hat{c}}^{(k)}, \boldsymbol{\Theta}_{\mathrm{c}}^{(k)})) > \epsilon_{\mathrm{dist}}\ (\forall c \neq \hat{c})$ **then**
    $p^{(k+1)}(\hat{c}) \leftarrow p^{(k)}(\hat{c}) + \alpha$
    $p^{(k+1)}(c) \leftarrow p^{(k+1)}(c) / \sum_c (p^{(k+1)}(c))$
  end if
end for

temporarily separable patterns. The reason is that if the sEMG signals of different classes are far apart, the misclassification rate is reduced since there are no overlapping patterns in the sEMG channel space at that time step. Thus, the classified result is considered reliable if the likelihood probability distribution of the predicted class is separable from other classes' probability distributions. The separability between Gaussian distributions is calculated using statistical distance $\mathsf{SD}(\boldsymbol{\Theta}_{\hat{\mathrm{c}}}^{(k)}, \boldsymbol{\Theta}_{\mathrm{c}}^{(k)})$, where $\boldsymbol{\Theta}_{\hat{\mathrm{c}}}^{(k)}$ and $\boldsymbol{\Theta}_{\mathrm{c}}^{(k)}$ are the parameters of Gaussian distributions, $\hat{\mathrm{c}}$ is the predicted class using the MAP classifier, and ϵ_{dist} is the reliability threshold. The reliability test condition is calculated as

$$\min_{\mathrm{c}} \mathsf{SD}\left(\boldsymbol{\Theta}_{\hat{\mathrm{c}}}^{(k)}, \boldsymbol{\Theta}_{\mathrm{c}}^{(k)}\right) > \epsilon_{\mathrm{dist}},\ \forall \mathrm{c} \neq \hat{\mathrm{c}}, \tag{8}$$

that checks whether the likelihood probability distribution of $\hat{\mathrm{c}}$ is sufficiently separable from other distributions. In this study, the Hellinger distance [11] was used as a statistical distance for measuring separability, where $\mathsf{SD}(\boldsymbol{\Theta}_{\mathrm{c}_1}, \boldsymbol{\Theta}_{\mathrm{c}_2}) = 1$ indicates fully separated distributions and $\mathsf{SD}(\boldsymbol{\Theta}_{\mathrm{c}_1}, \boldsymbol{\Theta}_{\mathrm{c}_2}) = 0$ indicates fully overlapping distributions ($\epsilon_{\mathrm{dist}} \simeq 0.95$). If the classification result $\hat{\mathrm{c}}$ is considered reliable, it is stored in the prior distribution of the next time step $p^{(k+1)}(\hat{\mathrm{c}})$ with the reinforcing parameter α as

$$p^{(k+1)}(\hat{\mathrm{c}}) \leftarrow p^{(k)}(\hat{\mathrm{c}}) + \alpha. \tag{9}$$

The proposed post-processing starts from the onset of the sequential sEMG input and ends at the offset; thus, the prior distribution is initialized to a discrete uniform distribution. Subsequently, the prior distribution of time step $k+1$ is updated as (9) for each time step.

- *Self-adjustment (MAP):* Reliable classification results of past sequential inputs were reflected in the prior distribution. Thus, the likelihood value in the overlapping pattern can be adjusted by a prior distribution, which gives a bias to the reliable class in the history of the sequence as Fig. 4(c). Notably, Bayes' theorem was applied to adjust the unreliable result; therefore, the maximum a posteriori (MAP) value is used for classification as

$$\begin{aligned} p^{(k)}(\mathrm{c}|x^{(k)}) &\propto p^{(k)}(\mathrm{c})\, p^{(k)}(x^{(k)}|\mathrm{c}) \\ \hat{\mathrm{c}} &\leftarrow \arg\max_{\mathrm{c}}\ p^{(k)}(\mathrm{c}|x^{(k)}). \end{aligned} \tag{10}$$

C. Case 2: K-Nearest Neighbor (KNN)

1) sEMG Trajectory: KNN calculates the Euclidean distance between the trained data points and the input signal, and it then selects the class that has the closest distance with the input sEMG signal.

- *sEMG trajectory $\boldsymbol{\Theta}_{\mathrm{c}}$:* KNN can also be extended to the sEMG trajectory, as in the case of the sEMG trajectory for MLC, if we consider the sEMG signals near the kth time step when classifying the kth sEMG input signal. Thus, the sEMG trajectory for KNN is defined as the sEMG signals near the kth time step, as follows:

$$\boldsymbol{\Theta}_{\mathrm{c}}^{(k)} = \left[x_{\mathrm{c},1}^{(k_{\mathrm{s}})} \cdots x_{\mathrm{c},1}^{(k_{\mathrm{e}})} \cdots x_{\mathrm{c},\mathrm{i_e}}^{(k_{\mathrm{s}})} \cdots x_{\mathrm{c},\mathrm{i_e}}^{(k_{\mathrm{e}})}\right], \tag{11}$$

where $i_{\mathrm{e}} = |\mathrm{I_{TRN}}|$ indicates the last index of the sEMG sequence in the training dataset. The sEMG trajectory $\boldsymbol{\Theta}_{\mathrm{c}}^{(k)} \in \mathbb{R}^{\mathrm{N} \times (\mathrm{k_e} - \mathrm{k_s}) \cdot |\mathrm{I_{TRN}}|}$ in (11) represents sEMG signals in the train set near time step k. Then, the sEMG trajectory of KNN is represented as (5).

- *Classification using the sEMG trajectory:* The KNN classification using sEMG trajectory only considers the sEMG signals near a certain time step k when classifying the sEMG signal of the kth input from the onset. Thus, the predicted class $\hat{\mathrm{c}}$ that has the minimum Euclidean distance ($\mathsf{ED}(\cdot,\cdot)$) between the input signal $x^{(k)}$ and sEMG values in $\boldsymbol{\Theta}_{\mathrm{c}}^{(k)}$ is selected as

$$\hat{\mathrm{c}} \leftarrow \arg\min_{\mathrm{c}} \mathsf{ED}(x^{(k)}, \boldsymbol{\Theta}_{\mathrm{c}}^{(k)}). \tag{12}$$

2) Post-Processing: The proposed post-processing for KNN uses Euclidean distance instead of statistical distance because KNN is a distance-based classifier.

- *Reliability test:* The reliability test for KNN has the same structure as MLC in (8); however, KNN is not a probabilistic method. Thus, the Euclidean distance was calculated instead of the statistical distance. $\min \mathsf{ED}(\boldsymbol{\Theta}_{\hat{c}}^{(k)}, \boldsymbol{\Theta}_{\mathrm{c}}^{(k)})$ calculates the minimum Euclidean distance between the elements of $\boldsymbol{\Theta}_{\hat{c}}^{(k)}$ and $\boldsymbol{\Theta}_{\mathrm{c}}^{(k)}$. Then, the reliability condition is calculated as follows:

$$\min_{c} \mathsf{ED}(\boldsymbol{\Theta}_{\hat{c}}^{(k)}, \boldsymbol{\Theta}_{\mathrm{c}}^{(k)}) > \epsilon_{\mathrm{dist}},\ (\forall c \neq \hat{c}). \tag{13}$$

This checks whether the sEMG train data labeled with the predicted class $\hat{c}$ are sufficiently separable from other classes of training data in the kth time step. If the result is considered reliable, the predicted class is stored in the kth element of the classification history array, $\mathcal{H}$ as

$$\mathcal{H}(k) \leftarrow \hat{c}. \tag{14}$$

In other words, $\mathcal{H}$ is an array of predicted classes with high reliability. The history array $\mathcal{H}$ for KNN replaces the role of the prior distribution for MLC.

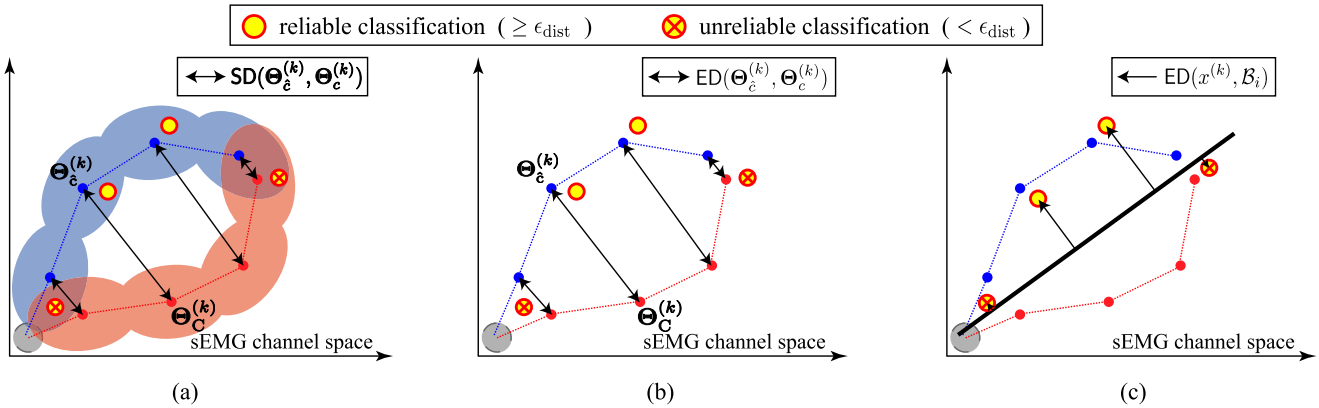

Fig. 5. sEMG trajectory is assigned only for MLC and KNN, with different parameters for each (probability distribution for MLC and sEMG data points for KNN). For the reliability test, MLC and KNN calculate the distance between two trajectories, whereas SVM calculates the distance from the decision boundary $\mathcal{B}_i$.

- *Self-adjustment:* If the result is not considered reliable, the modal class in the prediction history array $\mathcal{H}$ becomes the classification result of the current time step k as

$$\hat{c} \leftarrow \mathsf{mode}(\mathcal{H}), \tag{15}$$

where $\hat{c}$ in (15) is an adjusted result.

D. Case 3: Support Vector Machine (SVM)

This section provides a detailed implementation of the suggested method for an SVM (one versus one, linear kernel).

1) sEMG Trajectory: SVM need not be extended to the sEMG trajectory owing to its decision boundaries. The decision boundaries are determined by support vectors that often occur in overlapping patterns; therefore, transient sEMG inputs are classified well because they are far from the decision boundaries, as shown in Fig. 5. For the same reason, the reliability of the sEMG input can be verified using the distance between the sEMG input and the trained decision boundaries. The sEMG input, which is far from the decision boundaries, can be considered a reliable classification result because it is far from the support vectors, which indicate overlapping patterns. Thus, the sEMG trajectory is insignificant, and classification is conducted using the conventional SVM.

2) Post-Processing: The proposed post-processing for SVM uses Euclidean distance similar to KNN.

- *Reliability test:* As mentioned earlier, the reliable classification result is far from all decision boundaries. Thus, the classification result of $x^{(k)}$ is reliable if the minimum distance from all trained decision boundaries exceeds a threshold ϵ_{dist}, as

$$\min_{\mathcal{B}_i \in \mathcal{B}} \mathsf{ED}(x^{(k)}, \mathcal{B}_i) > \epsilon_{\text{dist}}. \tag{16}$$

We denote a set of decision boundaries as $\mathcal{B}$ and its element as $\mathcal{B}_i$. Subsequently, the reliable result is stored in the classification history set $\mathcal{H}$ as in (14).

- *Self-adjustment:* The adjustment process is similar to KNN in (15).

The overall structure of the proposed method is shown in Fig. 3, even if the corresponding classifier is changed; the correspondence is summarized in Fig. 5 and Table I .

E. Analysis Metric

1) Degree of Overlap in the Steady State Patterns: The metric for pattern overlapping can be calculated using the statistical distance; however, the value of the statistical distance does not intuitively show how much the signals overlap each other. In other words, although the statistical distance indicates whether the two distributions are different, the numerical value of the statistical distance does not quantify the area or volume of the overlapping patterns. Thus, the degree of overlap $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ is suggested to estimate the extent to which the steady signals of the two different classes C_1 and C_2 overlap. First, the class boundary MD_{C} was introduced to define the overlapped status by representing boundaries of the pattern occupied by each class's steady signals. Subsequently, $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ calculation is conducted.

- *Class boundary* MD_{C}*:* The class boundary is determined using the Mahalanobis distance and outlier detection with the interquartile range (IQR) method. We defined the set of steady state signals that are labeled with class C as

$$\mathcal{S}_{\mathrm{C}} = \left\{ x_{\mathrm{C}}^{(k)} \mid k > k_{\text{ts}} \right\}, \tag{17}$$

where a certain time step k_{ts} was defined to divide the transient and steady states as Fig. 2(a). The time index k_{ts} was defined manually as the maximum time step among the sEMG sequences when the amplitude of the transient signal after its peak became similar to that of the steady signal. Separating transient and steady states with a single timing seems unreasonable because the specific boundary between the two states is ambiguous, even if the shape of the transient signals differs from that of the steady signals. Considering this factor, we selected the maximum value of the time step k_{ts} to ensure that the signals of all sequences after k_{ts} have the characteristics of a steady signal. The

TABLE I
CORRESPONDING RELATIONSHIP BETWEEN CALCULATION METHODS OF EACH BASE CLASSIFIER

Classifier type	Classifier name	Reliability test	Self-adjustment
Probabilistic	MLC	Statistical distance between sEMG trajectories $\min_c \mathsf{SD}(\boldsymbol{\Theta}_{\hat{c}}^{(k)}, \boldsymbol{\Theta}_c^{(k)}) > \epsilon_{\text{dist}}$	Bayesian (Prior distribution)
Distance-based (Euclidean)	KNN	Euclidean distance between sEMG trajectories $\min_c \mathsf{ED}(\boldsymbol{\Theta}_{\hat{c}}^{(k)}, \boldsymbol{\Theta}_c^{(k)}) > \epsilon_{\text{dist}}$	Mode class (Stored reliable results)
	SVM	Euclidean distance from decision boundaries $\min_{\mathcal{B}_i \in \mathcal{B}} \mathsf{ED}(x^{(k)}, \mathcal{B}_i) > \epsilon_{\text{dist}}$	Mode class (Stored reliable results)

Mahalanobis distance $d_{\mathrm{M}}(x, \mathrm{Q_C})$ is calculated as

$$d_{\mathrm{M}}(x, Q_{\mathrm{C}}) = \sqrt{(x - \bar{\mathcal{S}}_{\mathrm{C}})\Sigma_{\mathcal{S}_{\mathrm{C}}}^{-1}(x - \bar{\mathcal{S}}_{\mathrm{C}})^{\top}}, \tag{18}$$

where a multivariate Gaussian distribution $\mathrm{Q_C}$ follows a mean $\bar{\mathcal{S}}_{\mathrm{C}}$ and covariance $\Sigma_{\mathcal{S}_{\mathrm{C}}}$. Subsequently, the Mahalanobis distance for each steady signal in class C is calculated. $\mathrm{MD}(\mathcal{S}_{\mathrm{C}}, \mathrm{Q_C})$ indicates the Mahalanobis distance between each steady signal in class C $(\mathrm{x} \in \mathcal{S}_{\mathrm{C}})$ and its distribution $\mathrm{Q_C}$ as

$$\mathrm{MD}(\mathcal{S}_{\mathrm{C}}, \mathrm{Q_C}) = \{\mathrm{d} \mid \mathrm{d} = \mathrm{d_M}(\mathrm{x}, \mathrm{C}), \mathrm{x} \in \mathcal{S}_{\mathrm{C}}\}. \tag{19}$$

The boundary of class C is defined using the outlier detection through the IQR method $(\text{outlier} > \mathrm{Q3} + 1.5 \times \mathrm{IQR})$; therefore, the boundary $(\mathrm{MD_C} \in \mathbb{R})$ is defined as

$$\mathrm{MD_C} = \mathrm{Q3}_{\mathrm{MD}(\mathcal{S}_{\mathrm{C}}, \mathrm{Q_C})} + 1.5 \times \mathrm{IQR}_{\mathrm{MD}(\mathcal{S}_{\mathrm{C}}, \mathrm{Q_C})}. \tag{20}$$

- *The degree of overlap* $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$*:* $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ calculates the ratio of the number of overlapping signals in class C_1, which overlap with class C_2, to the total number of signals in $\mathcal{S}_{\mathrm{C}_1}$. The overlapping signals indicate that the class C_1 signals were inside the boundary of class C_2. In other words, the Mahalanobis distance of $x \in \mathcal{S}_{\mathrm{C}_1}$ from distribution $\mathrm{Q}_{\mathrm{C}_2}$ was less than the boundary distance $\mathrm{MD}_{\mathrm{C}_2}$ of class C_2 as

$$\mathrm{d_M}(x, \mathrm{Q}_{\mathrm{C}_2}) \leq \mathrm{MD}_{\mathrm{C}_2}, \text{ where } x \in \mathcal{S}_{\mathrm{C}_1}, \tag{21}$$

and the overlapping signal is identified as

$$\delta_{\mathrm{C}_1,\mathrm{C}_2}(x) = \begin{cases} 1 & (d_{\mathrm{M}}(x, \mathrm{Q}_{\mathrm{C}_2}) \leq \mathrm{MD}_{\mathrm{C}_2}) \\ 0 & (d_{\mathrm{M}}(x, \mathrm{Q}_{\mathrm{C}_2}) > \mathrm{MD}_{\mathrm{C}_2}) \end{cases}, \tag{22}$$

which gives one if the signals overlap. Then, $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ is calculated as

$$\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2) = \frac{\sum_{x \in \mathcal{S}_{\mathrm{C}_1}} \delta_{\mathrm{C}_1,\mathrm{C}_2}(x)}{|\mathcal{S}_{\mathrm{C}_1}|}, \tag{23}$$

where $|\cdot|$ denotes the size of the set. The value of $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2) = 1$ indicates that all the class C_1 signals except outliers are inside the class boundary of class C_2, whereas $\mathrm{deg}(\mathrm{C}_1, \mathrm{C}_2) = 0$ indicates that they are outside the class C_2. Additionally, $\mathrm{degO}(\mathrm{C}_1)$ is introduced to express the degree of overlap for each motion as

$$\mathrm{degO}(\mathrm{C}_1) = \sum_{\substack{\mathrm{C}_2 \\ \mathrm{C}_1 \neq \mathrm{C}_2}} \frac{\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2) + \mathrm{degO}(\mathrm{C}_2, \mathrm{C}_1)}{2}, \tag{24}$$

and $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ and $\mathrm{degO}(\mathrm{C}_2, \mathrm{C}_1)$ was averaged because $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ is asymmetric. The reason for the asymmetry is explained in Section IV.

2) Classification Accuracy Calculation: We analyzed the classification accuracy of the transient and steady states. The prediction result at k^{th} time step is represented as

$$\delta_{\mathrm{i}}(k) = \begin{cases} 1 & (\hat{c}_{\mathrm{i}}^{(k)} = \boldsymbol{y}_{\mathrm{i}}^{(k)}) \\ 0 & (\hat{c}_{\mathrm{i}}^{(k)} \neq \boldsymbol{y}_{\mathrm{i}}^{(k)}) \end{cases}, \tag{25}$$

where the predicted class and true label are denoted as $\hat{c}_{\mathrm{i}}^{(k)}$ and $\boldsymbol{y}_{\mathrm{i}}^{(k)}$, respectively, for the kth time step sEMG input of the ith sequence in the test sEMG sequence $(i \in \mathrm{I_{TST}})$. $\mathrm{I_{TST}}$ represents the sEMG sequence index of the test dataset. The transient state signals vary with time; therefore, the transient state accuracy was analyzed for each time step as

$$acc_{inst}(k) = \frac{\sum_{i \in \mathrm{I_{TST}}} \delta_i(k)}{|\mathrm{I_{TST}}|} \quad (k < k_{\mathrm{ts}}). \tag{26}$$

We denote the accuracy in (26) as *instantaneous accuracy* because it represents the classification accuracy at a certain time step. The steady state accuracy is calculated as

$$acc_{\text{overall}} = \frac{\sum_{k \geq k_{\mathrm{ts}}} \sum_{i \in \mathrm{I_{TST}}} \delta_{\mathrm{i}}(k)}{|\mathrm{I_{TST}}| \cdot \sum_{k \geq k_{\mathrm{ts}}} 1}, \tag{27}$$

$$acc_{\mathrm{C}} = \frac{\sum_{k \geq k_{\mathrm{ts}}} \sum_{i \in \mathrm{I_{TST}}} \delta_{\mathrm{i}}(k)}{|\mathrm{I_{TST}}| \cdot \sum_{k \geq k_{\mathrm{ts}}} 1}, \text{ for } (\boldsymbol{y}_{\mathrm{i}}^{(k)} = \mathrm{C}), \tag{28}$$

where (27) represents overall classification accuracy and (28) represents the accuracy for each class.

III. EXPERIMENTAL SETUP

An experimental procedure is presented to verify the proposed method. This study was approved by the Institutional Review Board (IRB) of the Pohang University of Science and Technology (number PIRB-2022-E012). The sEMG signals were collected from 10 subjects (eight males, two females) using the Delsys Trigno Avanti system. Nine wireless sensor modules were used. Among them, eight sensor modules measured the

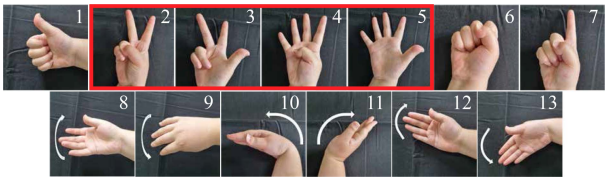

Fig. 6. Gestures of each class: The target case mostly appears in the gestures highlighted in the red box.

TABLE II
SPEARMAN'S RANK CORRELATION COEFFICIENT BETWEEN AVERAGE CLASSIFICATION ACCURACY AND ACCUMULATED $\mathrm{degO}(\mathrm{C}_1)$

Avg. accuracy	Accumulated $\mathrm{degO}(\mathrm{C}_1)$	p-value
MLC	-0.7088	0.007
KNN	-0.8242	0.001
SVM	-0.7802	0.002

sEMG signal at a sampling rate of 1259 Hz. The sensors were equidistantly placed on each subject's forearm. Each subject performed 13 types of hand gestures (Fig. 6) and repeated 15 times for each gesture. Experimental instructions were provided to the subjects with video and sound cues. The subjects performed one gesture for 4 s and rested for 4 s according to the instructions. Moreover, the subjects rested for at least 30 s after performing one type of gesture (15 repetitions) to avoid muscle fatigue. A sensor module measured the acceleration and angular velocity of the hand at a sampling rate of 148.1481 Hz. This sensor was attached to the back of the subject's hand to measure the onset and offset timing of the motion.

A bandpass filter (15–500 Hz) was used because of the frequency range of the sEMG signal, and a notch filter (60 Hz) was applied to reduce the surrounding noise. Subsequently, the MAV feature was applied to the filtered signal. A larger sliding window of the MAV feature causes a recognition delay in the prediction system; therefore, 100 ms windows were applied to take advantage of sEMG in gesture prediction. The onset and offset points of each gesture repetition were manually determined, as shown in Fig. 1(a). The first to fifth repetition signals for each gesture were used to make the sEMG trajectory (training), and the subsequent 10 repetitions were used to test classification accuracy.

IV. Results and Discussion

We assumed that the overlapping signal is correlated with the classification accuracy because the decision boundary is not easily determined between the overlapping class pairs. Thus, the correlation between the degree of overlap and classification accuracy is explained. Subsequently, the accuracy with and without applying the proposed post-processing was compared in both steady and transient states.

A. Degree of Overlap and Classification Accuracy

Fig. 7 shows the calculated result of the $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ using (23). If the color of $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ gets darker, most of the steady signals in class C_1 are found inside the pattern of class C_2, whereas a brighter color indicates that the two classes do not overlap. Classes 2 to 5 show dark colors, which indicate the high value of $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$. Thus, we conclude that the signals from those classes are quantitatively similar. The reason behind this is inferred from an anatomical factor. Classes 2 to 5 are finger extension motions, and the muscles responsible for the gesture are distributed in similar locations. Thus, the numerical values of these classes are quantitatively similar. The asymmetry of the $\mathrm{degO}(\mathrm{C}_1, \mathrm{C}_2)$ is shown in Fig. 7; the region where the class boundary forms is different for each class. For example, the result of subject 1 shows $\mathrm{degO}(1, 9) \simeq 0.76$ and $\mathrm{degO}(9, 1) \simeq 0$, which are asymmetric $(\mathrm{degO}(1, 9) \neq \mathrm{degO}(9, 1))$. These values indicate that the boundary of Class 9 is larger than that of Class 1; therefore, most of the Class 1's signals are inside the boundary of Class 9. Conversely, $\mathrm{degO}(9, 1)$ shows a low value because Class 1's signals occupy only a small part of the region that Class 9 has formed.

Fig. 8 describes the average classification accuracy of 10 subjects in steady state and the accumulated value of $\mathrm{degO}(\mathrm{C}_1)$ for each class. The average classification accuracy was obtained using three conventional classifiers without applying the suggested method. The accumulated value of $\mathrm{degO}(\mathrm{C}_1)$ was calculated by summing up the $\mathrm{degO}(\mathrm{C}_1)$ value for all subjects. Only the relative relationship between $\mathrm{degO}(\mathrm{C}_1)$ and average accuracy for each class was considered because we are interested in the correlation between classification accuracy and overlapping pattern. In other words, we cannot conclude that the value of the degree of overlap determines the classification accuracy because the degree of overlap only gives the possibility of misclassification.

A Spearman's correlation was calculated for each type of classifier to find out the relationship between the average classification accuracy and the accumulated degree of overlap $\mathrm{degO}(\mathrm{C}_1)$ for each class. A strong negative monotonic correlation between classification accuracy and accumulated $\mathrm{degO}(\mathrm{C}_1)$ was observed for MLC ($\mathrm{r_S} = -0.6978, \mathrm{n} = 13$), KNN ($\mathrm{r_S} = -0.7582, \mathrm{n} = 13$), and SVM ($\mathrm{r_S} = -0.7253, \mathrm{n} = 13$). The correlation was considered statistically significant ($p < 0.01$) as shown in Table II. The overlapping signals were negatively correlated with the classification accuracy and were included in our experiment.

B. Classification Accuracy in the Steady State

The performance of the proposed method was compared with three cases: without any post-processing, with single window majority voting (SWMV) [1], and with multi-window majority voting (MWMV) [15]. Fig. 9 shows the enhancement of the classification accuracy in the steady state for each subject when the post-processing is applied. The average accuracies of each classifier were 80.81% (MLC), 83.63% (KNN), and 88.02% (SVM) before implementing the post-processing. After its implementation, the average accuracy enhancement was the highest at approximately 8.12%p (MLC), 7.68%p (KNN), and 11.54%p (SVM) for the proposed method, achieving increased

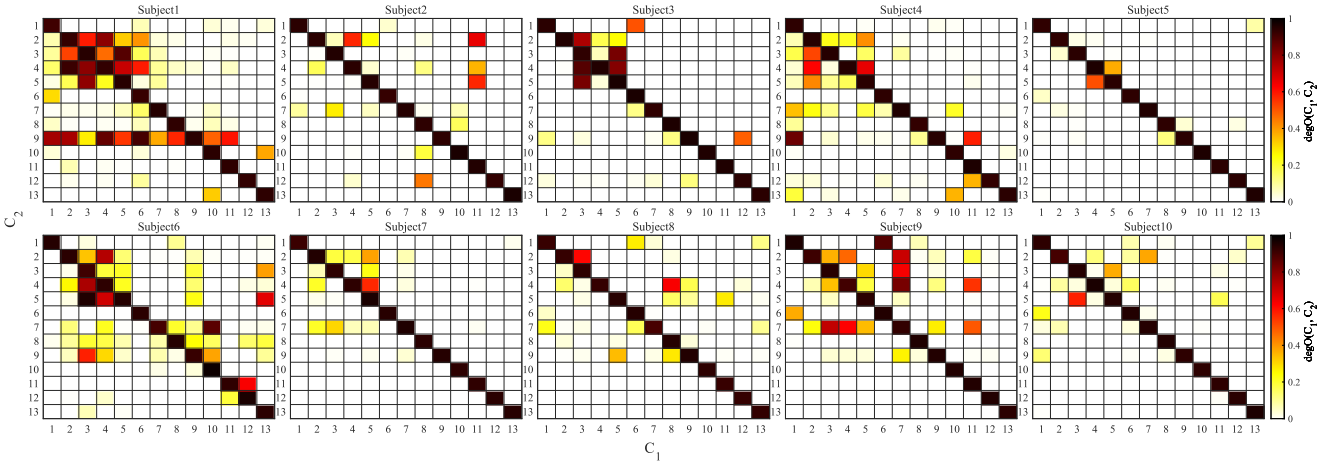

Fig. 7. Degree of overlap $\mathrm{degO}\,(\mathrm{C}_1, \mathrm{C}_2)$ for each subject.

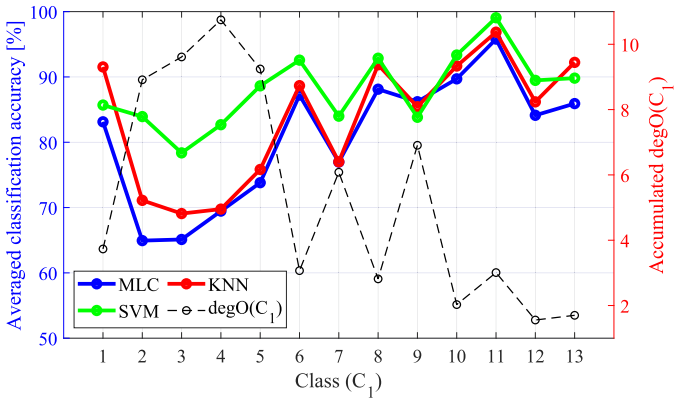

Fig. 8. Average classification accuracy and accumulated $\mathrm{degO}\,(\mathrm{C}_1)$ for each class.

average accuracies of 88.93% (MAP), 91.31% (proposed with KNN), and 99.56% (proposed with SVM). Therefore, the proposed method showed the highest accuracy improvement among other post-processing methods.

All post-processing methods reduced classification error; therefore, the majority voting strategy and the proposed method give robustness to gesture recognition performance. The accuracy of MWMV is higher than SWMV for most cases because MWMV exploits more sEMG information than SWMV strategy. However, classification results in the overlapping classes are erroneous; therefore, the voting majority may adjust the classification error with other errors while the voting process. One difference with the post-processing in the literature is that the proposed method uses reliable results from the separable patterns for adjusting error, so the adjusted result tends to become a correct classification.

Fig. 10 shows each class's classification accuracy of the steady state. The classification accuracy of each class was correlated with $\mathrm{degO}(\mathrm{C}_1)$, as explained in Section IV-A. The accuracy of each class increased for most of the classes, as shown in Fig. 10. In particular, Classes 2 to 5, which had a high degree of overlap, achieved +13.03%p (MLC), +13.42%p (KNN), and +15.90%p (SVM) enhancements in classification accuracy. Based on this result, the proposed method is considered to help decode the overlapping signal using past reliable classification results. Class 8, in the MLC-MAP comparison, shows a slightly lower classification accuracy (−1.37%p) because of the error-accumulation problem described below.

The proposed post-processing showed the highest performance; however, the enhancement rate differed slightly depending on the type of classifier. The difference between the probabilistic and distance-based classifiers is one reason for the different enhancement rates because the self-adjustment rule for each classifier creates a trade-off relationship in an error-accumulation problem. MLC uses Bayes' theorem to reflect reliable classification results in past sEMG inputs. If the value of the new sEMG sample almost coincides with the mean of the probability distribution of the incorrect class, the posterior probability of the incorrect class increases, regardless of how close to one is the prior probability of the correct class. However, the self-adjustment step in the proposed post-processing for KNN and SVM replaces the unreliable classification results in the current time step with a reliable result from the past time steps. In other words, the self-adjustment step does not reflect an unreliable classification result in the current time step, unlike the self-adjustment step of MLC. Thus, using the current value may degrade the prediction accuracy.

Conversely, the error can be accumulated when an incorrect prediction is made in the reliable region owing to the non-stochastic property of the sEMG. Thus, even if the correct class (but unreliable) is predicted in the overlapping patterns, the prediction result can be adjusted to the incorrect class because of the previously accumulated errors, which were considered reliable results. In rare cases, the likelihood values of the MLC classifier also compensate for the accumulated error of the prior distribution as the sEMG input approaches the mean of the correct class distribution; however, the likelihood values are also considered unreliable in the scope of our reliability condition.

Additionally, the difference in the enhancement rate was caused by the classification rule of the type of classifiers. MLC and KNN determine the class based on the closeness of the incoming signal to a class, and they identify whether the predicted class is reliable by comparing the separability of the sEMG

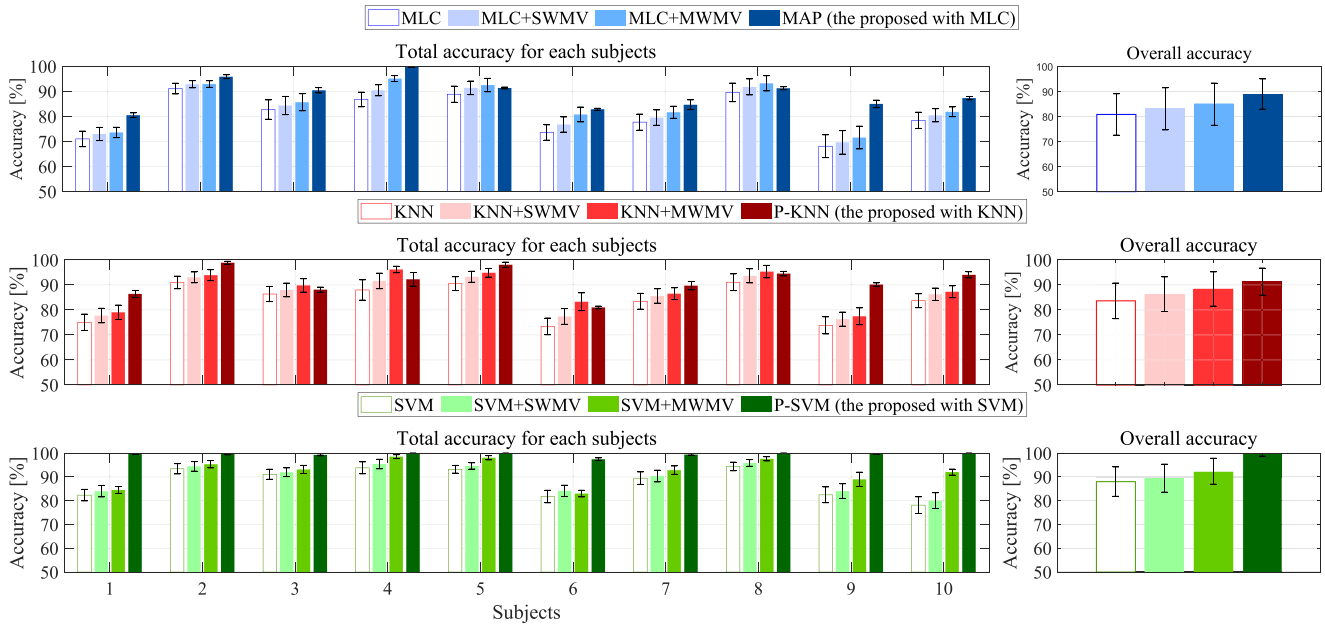

Fig. 9. Classification accuracy for each subject: Without post-processing, SWMV, MWMV, the proposed method.

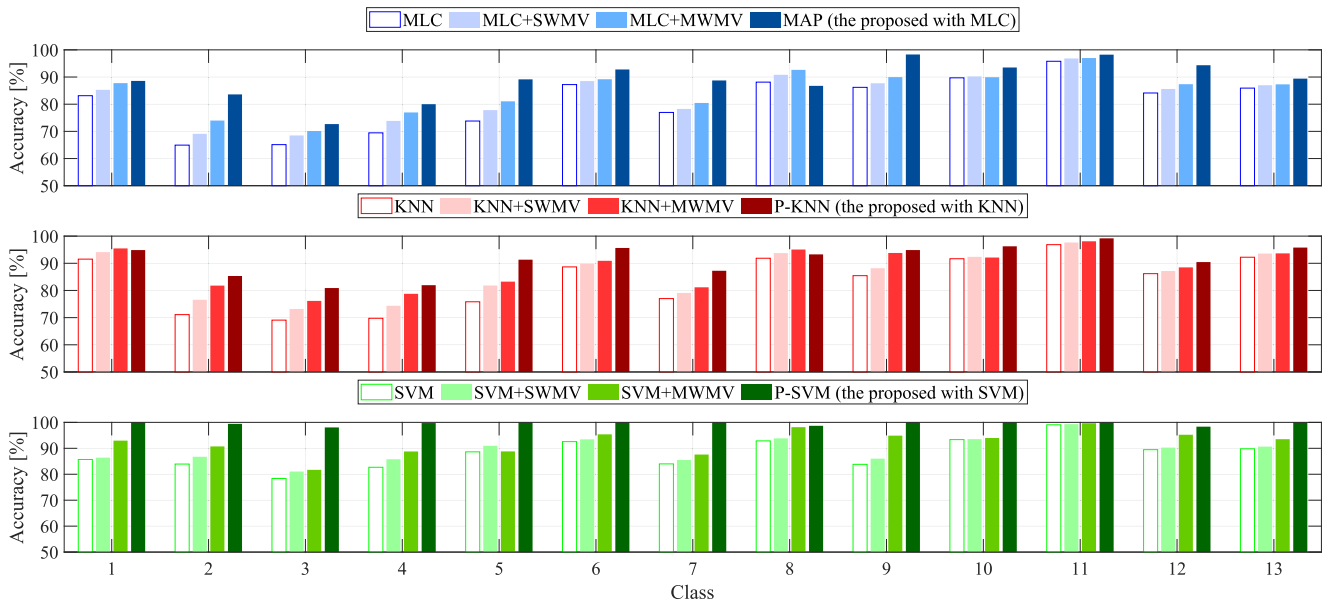

Fig. 10. Classification accuracy for each class: Without post-processing, SWMV, MWMV, the proposed method.

trajectory in the present time step. As explained above, these classifiers make incorrect predictions if the incoming signal approaches the incorrect class trajectories in a certain time step. However, SVM finds an optimal decision boundary that maximizes the margin and verifies the distance between the decision boundary and incoming signals. If the incoming signal is far from the decision boundary, it is a reliable classification result. Therefore, the classification accuracy tends to be high in SVM because of the optimal decision boundaries. Additionally, SVM does not suffer from the disadvantages of MLC and KNN; therefore, it can achieve a high accuracy enhancement rate when the proposed method is applied.

C. Classification Accuracy in the Transient State

An important reason for using sEMG for gesture recognition is that it can be detected before actual movements. Thus, it is important to test the accuracy depending on the time from the sEMG onset and gesture motion onset to offset. Fig. 11 shows the change in average classification accuracy over time (instantaneous accuracy) for ten subjects from the onset of the sequential sEMG data $(k = 1, \text{time} = 0)$ to the endpoint of the transient state $(k = k_{\text{ts}})$. The instantaneous accuracy was calculated using (26). The accuracy gradually increased with the time step of the sequence; therefore, the lowest accuracy was

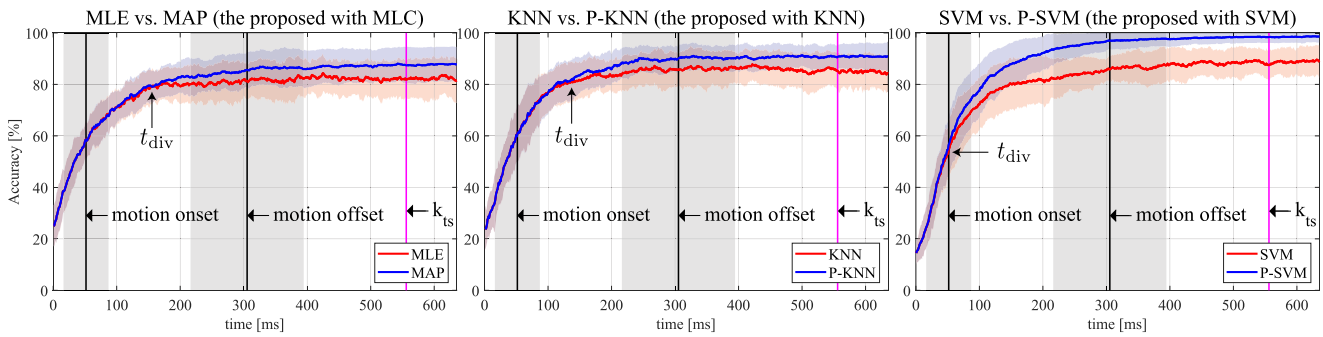

Fig. 11. Averaged instantaneous accuracy for 10 subjects in the transient state. Black lines indicate the onset and offset time of motion, and the light gray regions indicate the standard deviation of the onset and offset time. The colored line represents the accuracy, and the pale-colored region represents the standard deviation of the accuracy at that time step.

observed in the onset time step of the sEMG sequence. MLC and KNN started at approximately 25%, and SVM exhibited lower accuracy of approximately 15%. The accuracy of the onset time step had the lowest value because the signals of the onset point are rest state signals; therefore, they are not easily distinguished. The onset accuracies of MLC and KNN are higher than that of SVM because of the characteristics of the classifiers determining the class that has the closest distance to the incoming signal. Near the onset point of motion, the accuracy increased to 60% for the two classifiers and to 55% for the SVM.

The accuracy difference occurs because classification rules are different according to existing classifiers. MLC and KNN determine the class of input signal by choosing the nearest class to achieve the maximum probability value (MLC) or minimum distance (KNN); therefore, they provide the correct predicted class for certain sEMG inputs. Conversely, the slack variable of SVM causes the decision boundary to have low sensitivity to sEMG signals near the onset point. In other words, the sEMG signals near the onset do not contribute sufficiently to make decision boundaries; therefore, the decision boundary of SVM cannot easily deal with the signals near the onset point. However, the most important aspect in the analysis of the onset accuracy was that all three classifiers showed the lowest accuracies at the onset.

Subsequently, a division in classification accuracy at t_{div} appears between the classifier with and without implementing the proposed method. For two reasons, no difference between the two classification accuracy lines was observed before t_{div}. First, the classification results were unreliable, and the reliable results were not yet stored; therefore, the classification result could not be adjusted using reliable results in the past region. Otherwise, the classification results were reliable; thus, the proposed post-processing had little or no effect on classification. Thus, the two accuracy results are similar before the division.

The gap between the two averaged accuracy lines increases over time after the t_{div}, as shown in Fig. 11. The probability of predicting the correct class increases when performing the gesture, especially before the motion is completed. In other words, gesture recognition can be done faster than the gesture with a high probability of being correct. In detail, the accuracy at the motion offset is enhanced after the proposed method was implemented as shown in Table III (MAP (+4.23%p), P-KNN (+4.23%p), P-SVM (+11.24%p)); therefore, the gesture recognition performance before the motion completion was enhanced, and the same accuracy can be achieved in the faster time step. Additionally, the slightly higher accuracy of steady state compared with the accuracy of (k_{ts}) for the same classifier was observed, as shown in Table III. This is because the reliable region of some sEMG sequences appears after k_{ts}, thus making the classification accuracy slightly lower owing to the errors in some sequences before k_{ts}. Those errors are corrected if the reliable region is found in the steady state; hence, the accuracy slightly increases in the case of the implemented proposed method, whereas it decreases for the conventional classifiers.

TABLE III
AVERAGED CLASSIFICATION ACCURACY OVER TIME (AVG. ACCURACY% (STANDARD DEVIATION))

	motion offset	k_{ts}	steady state
MLC	81.46% (7.32)	81.77% (7.33)	80.81% (8.24)
MAP	85.69% (6.80)	87.38% (7.13)	88.93% (6.02)
KNN	85.85% (6.70)	85.31% (7.10)	83.63% (7.13)
P-KNN	90.08% (5.04)	90.69% (4.52)	91.31% (5.44)
SVM	86.38% (5.18)	88.08% (6.15)	88.02% (6.17)
P-SVM	97.62% (2.60)	99.08% (1.25)	99.56% (0.79)

The division in classification accuracy at t_{div} appears near the motion onset line in the case of SVM, approximately 50 ms after the motion onset time for MLC and KNN. A faster appearance of the division point (t_{div}) implies a faster occurrence of unreliable prediction results. This phenomenon is due to the characteristics of the SVM and the post-processing framework. The one-versus-one method was used in SVM, and it calculates the distance between the incoming sEMG sample and all the decision boundaries. Our method determines the minimum distance among the calculated distances and considers the sEMG sample unreliable if the minimum distance is smaller than the threshold. However, the incoming sEMG sample is more likely to be unreliable because all the distances between the decision boundaries should be large to obtain a reliable result. Thus, SVM has a faster appearance of division point. The overall accuracy enhancement rate was also the largest for the same reason. However, training SVM requires considerable time; therefore, a trade-off between accuracy and training time exists.

V. Conclusion

We proposed a post-processing method to enhance the performance of sEMG-based gesture recognition in the case of overlapping sEMG patterns. The main idea of this study is to use the temporal history (transient state) of the sEMG sequence because it behaves differently from steady state signals. In other words, overlapping patterns in the steady state can be separated when we observe patterns in the transient state. The proposed post-processing adjusts the predicted class in the overlapping pattern using a reliable classification history in the separated transient state. It is a post-processing method for classified results that can be implemented in conventional classifiers. Therefore, the implementation and performance enhancement of three classifiers (MLC, KNN, and SVM) were shown. Although only the MAV feature was used in this study, the proposed method can deal with other features if separable patterns in other feature spaces can be found in the temporal history of the sEMG feature sequence. The results demonstrated the advantages of observing the entire sEMG sequence instead of focusing only on the steady state when predicting gestures using the sEMG signal. The signals of the transient state can aid in decoding the overlapping signals in the steady state and allow fast classification even before the motion is completed. Future studies will focus on real-life applications because the post-processing deals with a single sEMG input for every time step; therefore, low computational complexity is expected. Onset and offset detection, including motion transition, is necessary to treat sEMG as sequential data, and real-time operation will be conducted for some applications, such as prostheses. Additionally, the effort to find proper parameters and train the sequence should be minimized to allow for developing a self-adaptive model for parameters and sEMG trajectory.

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